

Extension Principle

Fuzzy vector

Fuzzy vector is a vector containing fuzzy membership values.

If there is a fuzzy set A defined on universe of discourse $X = \{x_1, x_2, \dots, x_n\}$, then fuzzy vector

$$\begin{aligned} \mathbf{a} &= \{\mu_A(x_1), \mu_A(x_2), \dots, \mu_A(x_n)\} \\ &= \{\mu_A(x_i) \mid 1 \leq i \leq n\} \end{aligned}$$

Functions of fuzzy sets

Let us consider two fuzzy variables A and B defined on universe of discourse X and Y respectively, and a function $B = f(A)$.

$$B = \sum_{y \in Y} \frac{\max_{x=f^{-1}(y)} \mu_A(x)}{y}$$

Image set of A can also be found by using composition if the relation $R(A, B)$ is given.

$$\begin{aligned} B &= A \circ R \\ \text{or} \\ \mathbf{b} &= \mathbf{a} \circ R \end{aligned}$$

Zadeh's extension principle

In general, let us consider a function on input fuzzy variables $A_1, A_2, \dots, A_n$ defined on universe of discourse $X_1, X_2, \dots, X_n$ respectively, and with a single output variable B defined on universe of discourse Y .

$$\begin{aligned} B &= f(A_1, A_2, \dots, A_n) \\ f &: P(X_1 \times X_2 \times \dots \times X_n) \rightarrow P(Y) \end{aligned}$$

The extension principle says that

$$B = \sum_{y \in Y} \frac{\max_{x=f^{-1}(y)} \min(\mu_{A_1}(x_1), \mu_{A_2}(x_2), \dots, \mu_{A_n}(x_n))}{y}$$

For continuous valued function, replace max operation with supremum, sup.

Fuzzy transform

In the case of $f : \underline{A} \rightarrow \underline{B}$, although the input and output are fuzzy, the function itself is crisp.

The mapping of a crisp input into a fuzzy output is called fuzzy transform.

$$\underline{f} : \underline{A} \rightarrow \underline{B}$$

Even if the input is crisp, the output will be fuzzy because the function itself is not crisp.

If X and Y are finite, then a fuzzy transform f can be represented as a fuzzy relation R , or in the form of a matrix:

$$\tilde{R} = \begin{matrix} & y_1 & y_2 & \dots & y_j & \dots & y_m \\ \begin{matrix} x_1 \\ x_2 \\ x_i \\ x_n \end{matrix} & \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1j} & \dots & r_{1m} \\ r_{21} & r_{22} & \dots & r_{2j} & \dots & r_{2m} \\ r_{i1} & r_{i2} & \dots & r_{ij} & \dots & r_{im} \\ r_{n1} & r_{n2} & \dots & r_{nj} & \dots & r_{nm} \end{bmatrix} \end{matrix}$$

For a single input x_i , the output will be $B_i = f(x_i)$

$$B_i = \sum_{j=1}^m \frac{\mu_R(i, j)}{y_j}$$

or

$$b_i = \{r_{i1}, r_{i2}, \dots, r_{im}\}$$

Hence, the fuzzy image of the element x_i is given by the i^{th} row of the relation matrix R .

To generalize further, let us consider that the input is also fuzzy, i.e. $\underline{B} = \underline{f}(\underline{A})$. In this case, we can use the extension principle.

$$\mu_B(y) = \max_{x \in X} \left(\min(\mu_A(x), \mu_R(x, y)) \right)$$

or

$$b_j = \max_i (\min(a_i, r_{ij}))$$

One-to-one mapping

If there is a one-to-one crisp mapping between $x \subseteq X$ and $y \subseteq Y$, then for two fuzzy sets A and B with universe of discourse X and Y respectively,

$$\begin{aligned} B = f(A) &= f\left(\frac{\mu_A(x_1)}{x_1} + \frac{\mu_A(x_2)}{x_2} + \dots + \frac{\mu_A(x_n)}{x_n}\right) \\ &= \frac{\mu_A(x_1)}{f(x_1)} + \frac{\mu_A(x_2)}{f(x_2)} + \dots + \frac{\mu_A(x_n)}{f(x_n)} \end{aligned}$$

Mapping of cartesian products

Consider two universe of discourse X and Y . We need to find $B = f(X, Y)$.

If the mapping is one-to-one, then we can use a similar equation as in the case of one-to-one mapping between two sets.

$$B = f(X \times Y) = \left\{ \frac{\min(x_i, y_j)}{f(x_i, y_j)} \mid \forall x_i \in X, y_j \in Y \right\}$$

But, if the mapping is not one-to-one, then we have to use the extension principle.

$$\mu_B(z) = \left\{ \max_{z=f(x_i, y_j)} \min(\mu_X(x_i), \mu_Y(y_j)) \mid \forall x_i \in X, y_j \in Y \right\}$$